

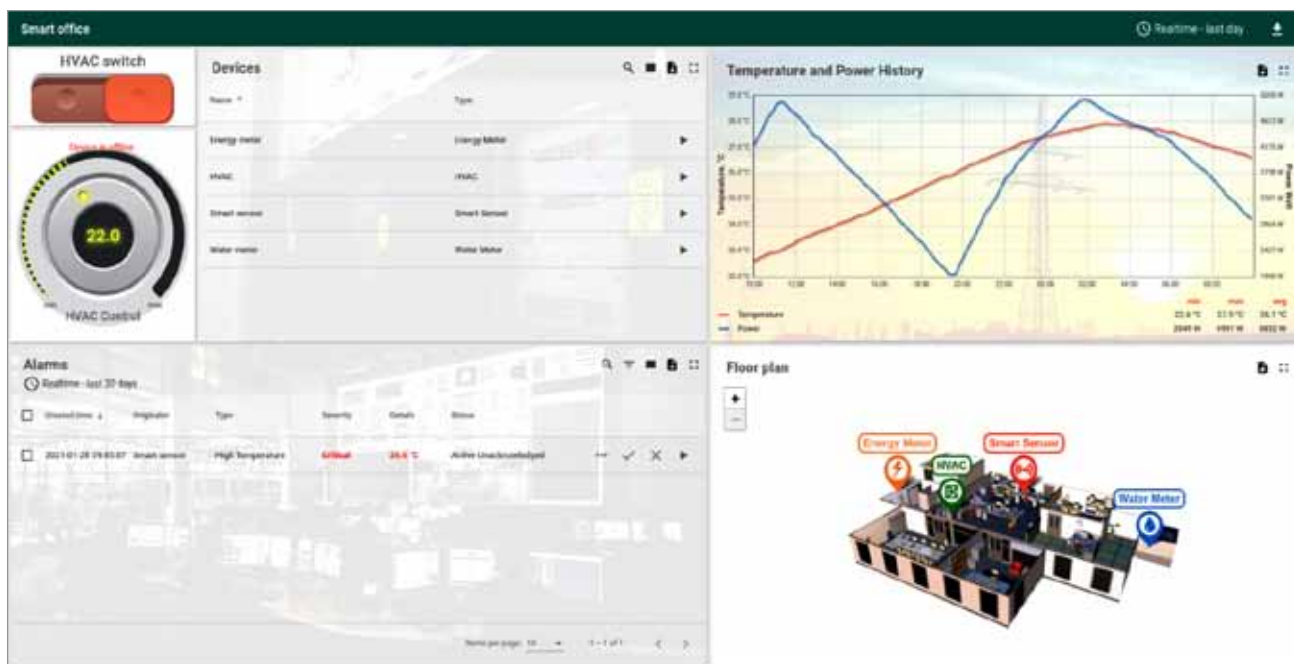


Fig. 3: Smart office dashboard

buildings, meeting rooms, equipment, devices, and data. Smart wristbands could become ID badge replacements (or smart ID badges could be created) in the office space. For example, the building will recognise your car when you arrive, authenticate it, and then direct you to a parking spot, thus providing a seamless user experience. Networked cameras can stream live video straight to central control, computer, and mobile devices. They help to:

- Monitor remote locations anywhere
- Give office access to staff remotely in real time
- Monitor real-time occupancy data

Meeting room/workplace reservation: This enables occupants to simplify workplace meeting room reservations. Instead of their wasting time looking for places to work and conduct meetings, a smart workspace can proactively carry out such routine work. Employees can use a digital workspace mobile app to book rooms and seats. Integration

with employees' calendars enables the system to send smart prompts to devices. This helps to:

- Book a room anywhere and in real time
- Sense when a room is occupied
- Know when a room is available
- Eliminate ghost meetings
- Find the right room to fit the needs of every meeting

With sensor technology, an individual doesn't have to manually sign in or out with their card, thus providing a seamless experience. Sensors will recognise when a person enters or exits a room, which can assist in knowing when meeting rooms are in use. In case people don't show up for a meeting or forget to cancel it (which is common with recurring meetings), these sensors can automatically set the meeting rooms free to be booked by other employees.

Wellness: Implementing wearable programs to increase employee health, wellness, and productivity helps to:

- Promote healthy habits
- Gamify wellness for employees

(making participation a fun and engaging 'game')

- Give employees benefits for participation

Eighty per cent of employees using healthy wellness programs report being more engaged at work.

Wayfinding and people finding:

Knowing the location of a person or thing unlocks many contextual digital workplace opportunities; examples include asset tracking, improving worker safety, understanding employee behaviour, predicting needs, and providing new services in the smart office.

Open source intelligent office solutions

There are many open source platforms available that can be used to build intelligent office solutions. Mainflux, ThingsBoard, DeviceHive, Thinger, Zetta, OpenRemote, Kaa, Fiware, ThingsConnect, WS02Rap are popular open source IoT platforms.

ThingsBoard: This platform dramatically reduces the effort needed to create smart office solutions. It is